TOTAL COST OF OWNERSHIP (TCO)

COMPREHENSIVE FINANCIAL ANALYSIS REPORT

Project: p2

Generated: December 27, 2025

EXECUTIVE SUMMARY

Initial Investment: €481,000.00

Annual Operating Costs: €512,640.00

5-Year Total Cost: €3,043,200.00

ROI: 3.4%

Payback Period: 1.4 years

1. INITIAL INVESTMENT BREAKDOWN

Equipment Purchase:	**€280,000.00**
Installation:	**€56,000.00**
Infrastructure:	**€120,000.00**
Training:	**€15,000.00**
Initial Consulting:	**€10,000.00**
TOTAL INITIAL INVESTMENT	**€481,000.00**

2. ANNUAL OPERATING COSTS

Energy:	**€24,240.00**
Maintenance:	**€11,200.00**
Labor:	**€180,000.00**
Insurance:	**€4,200.00**
Consumables:	**€285,000.00**
Utilities:	**€8,000.00**

TOTAL ANNUAL OPERATING COSTS	**€512,640.00**

3. ASSET-BY-ASSET COST BREAKDOWN

SJSZ 65/132 Conical Twin-Screw Extruder

Supplier: SJSZ 65/132 Conical Twin-Screw Extruder Supplier 2

Purchase Price: €150,000.00

Installation: €30,000.00

Annual Maintenance: €6,000.00

Annual Energy: €17,760.00

Vacuum Calibration Table

Supplier: Vacuum Calibration Table Supplier 2

Purchase Price: €50,000.00

Installation: €10,000.00

Annual Maintenance: €2,000.00

Annual Energy: €4,080.00

Automatic Chipless Cutter

Supplier: Automatic Chipless Cutter Supplier 2

Purchase Price: €35,000.00

Installation: €7,000.00

Annual Maintenance: €1,400.00

Annual Energy: €960.00

Online Lamination Machine

Supplier: Online Lamination Machine Supplier 2

Purchase Price: €45,000.00

Installation: €9,000.00

Annual Maintenance: €1,800.00

Annual Energy: €1,440.00

4. FINANCIAL METRICS

5-Year Total Cost:	**€3,043,200.00**
10-Year Total Cost:	**€5,605,400.00**
NPV (5 years, 10% discount):	**€822,450.00**
NPV (10 years, 10% discount):	**€1,610,340.00**
IRR:	**0.68%**
ROI:	**3.35%**
Payback Period:	**€1.42years**

5. 10-YEAR FINANCIAL PROJECTIONS

Year	Revenue	Costs	Net Cash Flow	Cumulative
1	€850,000	€512,440	€337,560	€-143,440
2	€875,500	€527,813	€347,687	€204,247
3	€901,765	€543,647	€358,118	€562,365
4	€928,818	€559,956	€368,862	€931,227
5	€956,682	€576,755	€379,927	€1,311,154
6	€985,383	€594,058	€391,325	€1,702,479
7	€1,014,944	€611,880	€403,064	€2,105,543
8	€1,045,393	€630,236	€415,157	€2,520,700
9	€1,076,755	€649,143	€427,612	€2,948,312
10	€1,109,057	€668,617	€440,440	€3,388,752

6. RISK ANALYSIS

High Risk Factors:

• Volatility in PVC resin market prices

• Energy price fluctuations in the European market

• Regulatory changes regarding plastic manufacturing emissions

• Technical downtime of the primary extruder

Mitigation Strategies:

• Establish long-term fixed-price contracts with raw material suppliers

• Install energy-efficient LED lighting and power factor correction systems

• Implement a rigorous preventive maintenance schedule (TPM)

• Invest in secondary filtration for water cooling to extend pump life

Sensitivity Analysis:

• 10% increase in raw material costs: Reduction of annual net cash flow by approximately $28,500, extending payback by 3 months.

• 20% increase in energy costs: Annual operating cost increase of $4,848; minimal impact on overall IRR.

• Production downtime of 15%: Significant risk to NPV, potentially reducing 10-year valuation by $250,000.

7. CONCLUSION

This comprehensive TCO analysis shows a total investment of €481,000 with annual operating costs of €512,640. The project demonstrates a 3.4% ROI with a payback period of 1.4 years. The financial projections indicate positive cash flow and sustainable growth over the 10-year period.